

POLICY MEMO

Social Vulnerability Mapping for Flood Management in India

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To : Members of the National Disaster Management Authority of India (NDMA)

Subject : Social Vulnerability Mapping for Flood Management

Background

India faces the highest flood damage in Asia (1), with annual losses averaging **\$213 million**, affecting over **75 lakh hectares** and claiming around **1,600 lives** (2). Recent annual costs have escalated up to **\$1.7 billion** (3). Urbanisation has compounded the problem, with unplanned development encroaching on floodplains and aggravating urban floods (4). Socially vulnerable populations usually live in informal settlements located on these encroached lands, often characterised by open drains and sewers. As a result, they bear a disproportionately large burden of the aforementioned losses. In fact, India has the **largest number of poor people who are exposed to flood risk in the world** (5).

Despite significant investments in structural measures like dams and reservoirs, states have underinvested in non-structural measures such as floodplain zoning, **social vulnerability mapping**, and drainage inspections (4). The political appeal of tangible infrastructure projects overshadows the need for intangible measures like vulnerability mapping. Tamil Nadu's recent social vulnerability assessment is a notable exception, but efforts across the country remain inconsistent and fragmented (6). Moreover, without **comprehensive data** on socially vulnerable populations, existing flood management departments, such as Public Works Departments, and urban planning bodies, lack a coordinated approach towards aiding these populations. This results in **poorly targeted** pre and post-disaster management.

Policy Recommendation

To tackle this challenge, the NDMA must incentivise states to undertake social vulnerability mapping. This can be achieved through the creation of a **Central Open Access Database for Social Vulnerability Mapping** that integrates data from all states and makes it **accessible in real time**.

The proposed system would assign a **Social Vulnerability Index (SVI)** score to each district on a scale from 1 to 10, with 10 representing the highest vulnerability. The SVI score would be calculated based on key indicators such as physical and climatic conditions of the region, accessibility to emergency services, household resistance capacity indicators such as income, housing conditions, and individual resistance capacity indicators such as age, education, health conditions, etc. This data would be collected from primary household surveys, local government records, and census data. Individual household data will remain confidential, with public access limited to aggregated district-level SVI scores, to prevent privacy breaches.

The SVI scores would be visualised through a **GIS map**, with colour-coding that allows users—especially the public—to easily identify vulnerable regions. The map would be updated in real-time based on physical vulnerability and climatic changes, while socio-economic indicators would be updated annually for the most vulnerable districts and every 2-3 years for others.

The State Disaster Management Authorities (SDMAs) will be responsible for gathering the data, while the NDMA will oversee and maintain the centralised platform.

Expected Impact

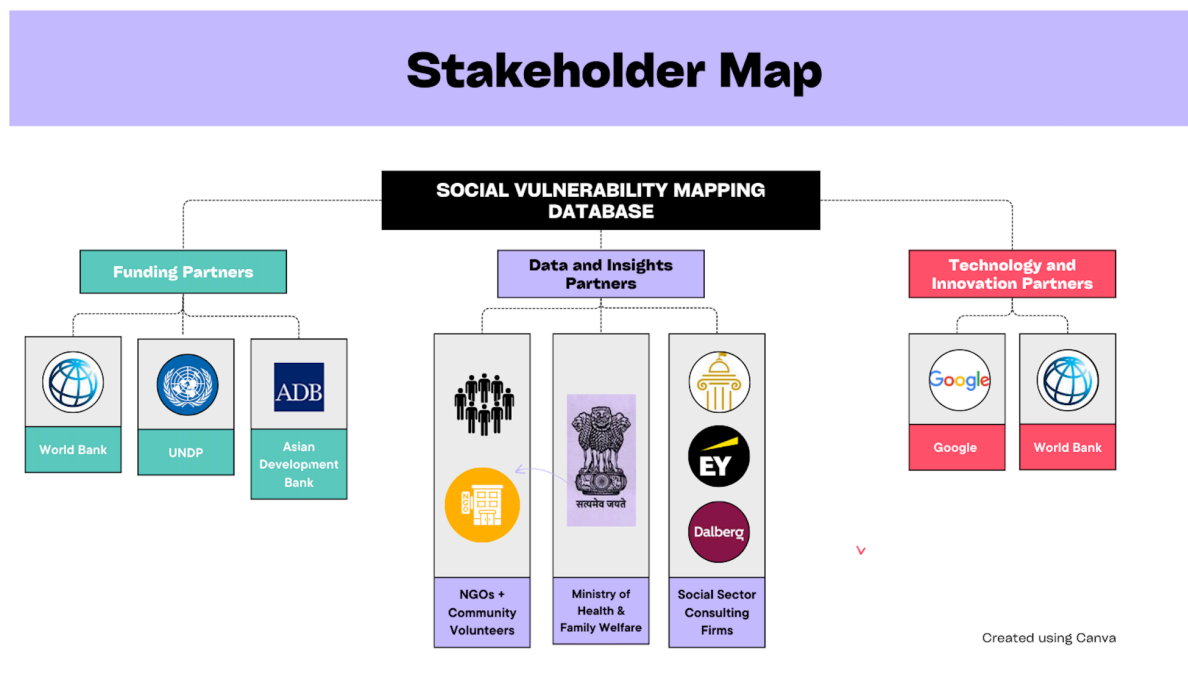
Social Vulnerability mapping has been undertaken in several countries ranging from the US and Australia to developing countries like Bangladesh. Studies show that integrating GIS methods to SVI enhances the understanding of factors increasing vulnerability and informs actions to be taken by respective authorities (7). The central open access database could thus help in the following ways :

- 1) **Political Incentives:** The public availability of SVI scores through GIS maps motivates state governments to improve their scores across dimensions like housing, education, healthcare, and flood preparedness, to increase political goodwill by depicting tangible developmental progress.
- 2) **Enhanced Coordination:** The multi-dimensional SVI framework necessitates collaboration among various government departments. Access to centralised, real-time data enables these departments to operate in Joint Work Groups, fostering cohesive

flood management plans. For example, land zoning authorities and flood forecasting teams can align efforts to prevent encroachments on floodplains.

- 3) **Involving Private Stakeholders:** Public access to information on vulnerable districts empowers private stakeholders such as NGOs and civil societies to target interventions effectively.

Implementation Plan:



International organisations such as the World Bank, UNDP, and ADB have funded similar projects globally. For example, the World Bank has contributed \$500 million to Bangladesh's social vulnerability mapping project (8). These funding bodies can ensure a steady financial foundation for the database.

NGOs and community volunteers can contribute towards data collection for household surveys (6). The **Ministry of Health and Family Welfare**, which annually releases rural household survey datasets (NFHS), can provide complementary data to enrich the SVI framework. Additionally, social sector consulting firms like **Samagra, Dalberg, and EY** can assist in the construction of the index, and in training volunteers.

Google's expertise in AI-based flood forecasting systems in India (9) and the **World Bank's** experience in both GIS mapping and social vulnerability assessment projects globally, positions them as ideal technical partners.

Budget

On analysis of similar projects by the Federal Emergency Management Agency (FEMA) in the US, the budget for the setup of this project is approximately **\$40 million** (10). With 50% funding from International Organisations, it will cost the NDMA **\$20 million** which is only **1.5%** of the budget of **\$1355 million** allocated to long term flood management in India (11).

Timeline



1) Planning and Data Collection includes:

- Finalising dimensions of the SVI
- Standardising data collection protocols across states
- Conducting household surveys and integrating this with existing government data.

2) **Development and Rollout involves:**

- a) Creating the central database,
- b) Mapping data to a GIS visualisation
- c) Implementing data privacy safeguards for household data
- d) Phased rollout, prioritising high-risk states, to continuously test and refine the platform.

Potential challenges include data manipulation, which will be managed by having NDMA coordinators verify state data before updates.

3) **Training and Capacity Building** of the NDMA and SDMA staff will begin in the early stages and extend into this phase, focusing on:

- a) Database management
- b) Data analysis and creation of reports to aid policymaking
- c) Public awareness campaigns

Conclusion

The first step towards effective disaster management for vulnerable populations is recognising who this population constitutes and what exactly they require. This data-driven platform seeks to provide the concerned authorities with this information while also incentivising them to coordinate and take measurable actions towards better flood resilience and restoration.

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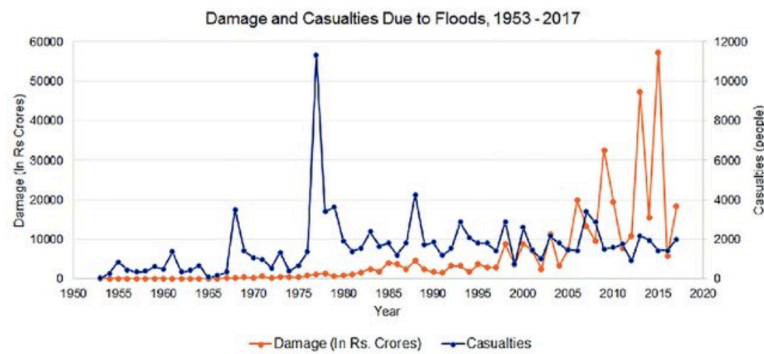
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APPENDIX

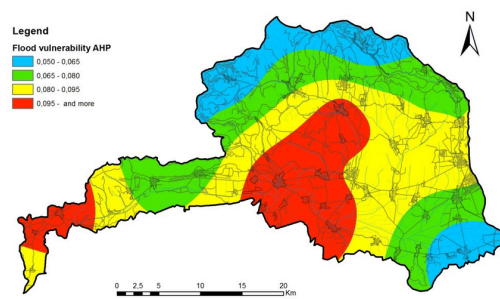
1. Graph Depicting Flood Damage in India



Source:

https://www.researchgate.net/publication/342736565_Integration_of_SAR_and_multi-spectral_imagery_in_flood_inundation_mapping-a_case_study_on_Kerala_floods_2018/figures?lo=1

2. Example of a GIS Map showing Flood Vulnerability



3. Budget Breakdown :

Data Collection	\$10 Million
Database Development	\$22 Million
Training and Capacity Building	\$8 Million
Total	\$40 Million

Calculated using figures from the FEMA budget. The FEMA figures were inflated by a factor of 1.5 to account for the larger population of India.